

DOCUMENT NAME: FORA Communication Protocol for SPO2 series	
D/N:	Rev:
FILE NAME: SPO2	Update: 2015/11/06

Revision History

#	Date.	Done by.	The update.	New rev.
1	2011/11/15	Aho	Primary Edition.	Ver1.0
2	2012/05/14	Carol	Modification.	Ver1.01
3	2012/09/21	Carol	Modification.	Ver1.02
4	2013/10/25	Carol	Modification.	Ver1.03
5	2015/04/23	Carol	Modification.	Ver1.04
6	2015/11/06	Carol	Note for different Bluetooth versions.	Ver1.05

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Table Of Contents

1. GENERAL.....	3
1.1. OVERVIEW	3
1.2. DOCUMENT OVERVIEW	3
1.3. DEFINITIONS, ACRONYMS, AND ABBREVIATIONS	4
2. INTERFACE SPECIFICATION.....	5
2.1. FRAME STRUCTURE.....	5
2.2. COMMAND LIST	7
2.2.1. [0x23] READ DEVICE CLOCK TIME (4 BYTES)	8
2.2.2. [0x24] READ DEVICE MODEL	9
2.2.3. [0x25] READ THE STORAGE DATA WITH INDEX, PART 1 (TIME).....	10
2.2.4. [0x26] READ THE STORAGE DATA WITH INDEX, PART 2 (RESULT)	11
2.2.5. [0x27] READ DEVICE SERIAL NUMBER, PART 1	12
2.2.6. [0x28] READ DEVICE SERIAL NUMBER, PART 2	13
2.2.7. [0x2B] READ STORAGE NUMBER OF DATA	14
2.2.8. [0x33] WRITE SYSTEM CLOCK TIME (4 BYTES)	15
2.2.9. [0x49] READ REAL TIME DATA.....	16
2.2.10. [0x50] TURN OFF THE DEVICE	16
2.2.11. [0x52] CLEAR/DELETE ALL MEMORY	17

1. General

1.1. Overview

The document describes the commands which will be used to communicate with spo2 meter.

Before retrieving data in the device, please make sure you have already established the proper connection channel between the device and the host side. For example, if you use USB cable, be sure that the USB driver has been properly installed; or if you use Bluetooth communication, be sure that the connection has been successfully established.

Generally speaking, the device will be the slave side waiting for the request from the host which is usually the computer or the hub/gateway.

Note for different Bluetooth versions:

For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by for both pairing and connection, the PIN for pairing is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired before any connection, please find the instruction in meter manual for the pairing process. The meter supports SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:

UUID Base: 1212-efde-1523-785feabcd123

Service UUID: 0x1523

Characteristic: 0x1524 (write/notify)

1.2. Document Overview

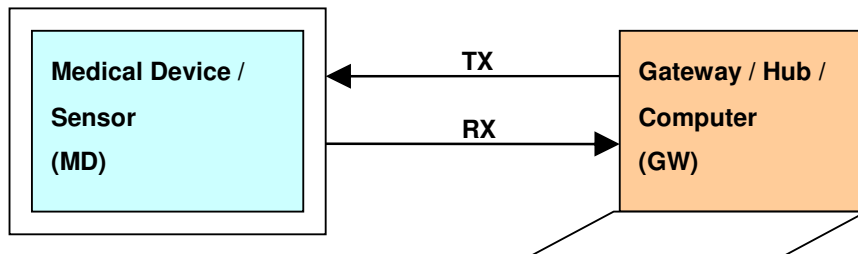
For the first time user on the device integration, please read the 2.1 frame structure. If you are familiar with serial port communication, you can review the command list table in section 2.2 directly. The following sub sections are all about the details for each command.

1.3. Definitions, Acronyms, and Abbreviations

GW	Gateway / Hub / Computer side
MD	SPO2 meter.
CMD	Command. Usually it's a one byte alphanumeric symbol.
ACK	Acknowledgement. Usually it's a one byte alphanumeric symbol.
NA	Not available

2. Interface specification

Block Diagram



The communication interface is base on RS232 serial port communication :

Baud rate	19200
Data bits	8
Parity	No
Start bit	1
Stop bit	1

2.1. Frame Structure

Byte	1	2	3 ~ n-2	n-1	n
Name	Start	CMD	Index, Address, or Data	Stop	Check Sum
Format	HEX				
GW→MD	51	CMD		A3	[1...n-1]
GW←MD	51	ACK		A5	[1...n-1]

ie. Check sum means the summation from byte 1 to byte n-1 with 8-bit length

Example : 8-byte frame

Byte	1	2	3	4	5	6	7	8
Name	Start	CMD	Data_0	Data_1	Data_2	Data_3	Stop	ChkSum

Format	HEX							
GW→MD	51	CMD	Data_0	Data_1	Data_2	Data_3	A3	[1..7]
GW←MD	51	ACK	Data_0	Data_1	Data_2	Data_3	A5	[1..7]

ie. Check sum means thefrom byte 1 to byte 7 with 8-bit length

2.2. Command List

Message name	CMD/ACK	Direction	Length (Bytes)	Note
Read device clock time	0x23	GW→MD	8	
	0x23	GW←MD	8	
Read device model.	0x24	GW→MD	8	
	0x24	GW←MD	8	
Read storage data, part 1(time)	0x25	GW→MD	8	
	0x25	GW←MD	8	
Read storage data, part 2(result)	0x26	GW→MD	8	
	0x26	GW←MD	8	
Read device serial number, part 1	0x27	GW→MD	8	
	0x27	GW←MD	8	
Read device serial number, part 2	0x28	GW→MD	8	
	0x28	GW←MD	8	
Read storage number of data	0x2B	GW→MD	8	
	0x2B	GW←MD	8	
Write device clock time	0x33	GW→MD	8	
	0x33	GW←MD	8	
Read real time data	0x49	GW→MD	8	
	0x49	GW←MD	8	
Force Slave Device goes to Sleep (Turn OFF)	0x50	GW→MD	8	
	0x50	GW←MD	8	
Clear/Delete all memory	0x52	GW→MD	8	
	0x52	GW←MD	8	

2.2.1. [0x23] Read device clock time (4 bytes)

Message Name:	Read device clock time (4 bytes)
Message ID:	0x23
Message Description:	Read device date and time (to minute)
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x23	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x23	Day + Month + Year		Minute	Hour

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x23	GW commands MD to response device clock time
	DATA	4	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x23	MD response device clock time
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word) (see Table A)
	DATA	2		Data_2 : Minute, Data_3 : Hour (see Table B)

Table A: Day + Month + Year

MSB								LSB							
Data_1								Data_0							
Year (7-bit)							Month (4-bit)				Day (5-bit)				
6	5	4	3	2	1	0	3	2	1	0	4	3	2	1	0

Table B: Minute and Hour

MSB								LSB							
Data_3								Data_2							
NA			Hour (5-bit)					NA		Minute (6-bit)					
			4	3	2	1	0			5	4	3	2	1	0

2.2.2. [0x24] Read device model

Message Name:	Read device model
Message ID:	0x24
Message Description:	Read device model / project code
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x24	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x24	Project Code			

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x24	GW commands MD to response device model
	DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x24	MD response project code
	DATA	2		Data_1 + Data_0 : Device model in four digits (Data_1 is the

			higher byte)
DATA	2		Data_3, Data_2 : undefined

2.2.3. [0x25] Read the storage data with index, part 1(time)

Message Name:	Read the storage data with index, part 1(time). Not all models are supported. PO100 is not supported.
Message ID:	0x25
Message Description:	Read data (part 1: measure date and time)
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x25	Index		0x0	0x0
Response from device.	GW←MD	0x25	D_M_Year		Minute	Hour

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x25	GW commands MD to response measure data (part 1)
	Index	2		0x00 is the latest reading in the device

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x25	MD response measure date and time
	D_M_Year	2		The measure date (see Table A in 2.2.1 [0x23])
	Minute	1		The measure minute (see Table B)
	Hour	1		The measure hour (see Table B)

2.2.4. [0x26] Read the storage data with index, part 2(result)

Message Name:	Read the storage data with index, part 2(result). Not all models are supported. PO100 is not supported.
Message ID:	0x26
Message Description:	Read data (part 2: measure result)
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x26	Index		0x0	0x0
Response from device.	GW←MD	0x26	SpO2		0x0	Heart Rate

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x26	GW commands MD to response measure data (part 1)
	Index	2		0x00 is the latest reading in the device

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x26	MD response measure date and time
	SpO2	2		The measure result (see Table C below) The unit is %
	Heart Rate	1		The measure result (see Table C below) The unit is BMP(pul/min)

Table C: SpO2

MSB	LSB
Data_1	Data_0
SpO2	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Table D: Heart rate

MSB								LSB							
Data_3								Data_2							
Pulse (8-bit) (Unit= pul/min)								N/A							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0

2.2.5. [0x27] Read device serial number, part 1

Message Name:	Read the device serial number, part 1
Message ID:	0x27
Message Description:	Read the device serial number, part 1. This command cooperates with 0x28 to complete the serial number.
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + ... + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read device serial number, part 1	GW→MD	0x27	0x0	0x0	0x0	0x0
Response from device	GW←MD	0x27	SN_0	SN_1	SN_2	SN_3

Field Description

#	Field	Size (Bytes)	Value	Description.
---	-------	--------------	-------	--------------

CMD	1	0x27	GW commands MD to response the device serial number, part 1
DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x27	Read the device serial number, part 1
	DATA	4		Data_0 ~ Data_3 : SN_0 ~ SN_3

2.2.6. [0x28] Read device serial number, part 2

Message Name:	Read the device serial number, part 2
Message ID:	0x28
Message Description:	Read the device serial number, part 2. This command cooperates with 0x27 to complete the serial number.
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + ... + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read device serial number, part 2	GW→MD	0x28	0x0	0x0	0x0	0x0
Response from device	GW←MD	0x28	SN_4	SN_5	SN_6	SN_7

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x28	GW commands MD to response the device serial number,

			part 2
	DATA	4	0x0

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x28	Read the device serial number, part 2
	DATA	4		Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.7. [0x2B] Read storage number of data

Message Name:	Read storage number of data. Not all models are supported. PO100 is not supported.
Message ID:	0x2B
Message Description:	Read data counts
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read Storage Number of Stored Readings	GW→MD	0x2B	User	0x0	0x0	0x0
Response from device	GW←MD	0x2B	Storage Number	0x0	0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x2B	GW commands MD to response the Number of Stored Readings

#	Field	Size	Value	Description.
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		(Bytes)		
	ACK	1	0x2B	MD response the Number of Stored Readings
	DATA	2		Data_1+Data_0 = Storage Number (Word)

2.2.8. [0x33] Write System Clock time (4 bytes)

Message Name:	Write device clock time (4 bytes)
Message ID:	0x33
Message Description:	Write the device date and time (to minute)
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Please refer to 0x23 command section for the format of date and time.

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Write System Clock time	GW→MD	0x33	Day+Month+Year	Minute	Hour	
Response from device.	GW←MD	0x33	Day+Month+Year	Minute	Hour	

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x33	GW commands MD to write System Clock time (4 bytes)
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word)
	DATA	2		Data_2 : Minute, Data_3 : Hour

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x33	MD write System Clock time (4 bytes) O.K.
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word)
	DATA	2		Data_2 : Minute, Data_3 : Hour

2.2.9. [0x49] Read real time data

Message Name:	Read real time data
Message ID:	0x49
Message Description:	Read real time data
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read real time data	GW→MD	0x49	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x49	SpO2		0x0	Heart Rate

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x49	GW commands MD to response real time data
	Data	4		0x00

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x49	MD response real time measure data
	SpO2	2		The measure result (see Table C) The unit is %
	Heart Rate	1		The measure result (see Table D) The unit is BMP(pul/min)

2.2.10. [0x50] Turn off the device

Message Name:	Turn off the device
Message ID:	0x50
Message Description:	Turn off the device from GW

Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Turn off the device	GW→MD	0x50	0x0	0x0	0x0	0x0
	GW←MD	0x50	0x0	0x0	0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x50	GW commands MD to turn off the power.
	DATA	4	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x50	MD responses.
	DATA	4	0x0	NA

2.2.11. [0x52] Clear/Delete all Memory

Message Name:	Clear/Delete all memory. Not all models are supported. PO100 is not supported.
Message ID:	0x52
Message Description:	Delete all measurement data in the device
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
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Clear/Delete all Memory Storages	GW→MD	0x52	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x52	0x0	0x0	0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x52	GW commands MD to Clear/Delete all Memory Storages in Device EEPROM
	DATA	1		
	DATA	3	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x52	Clear/Delete all Memory Storages in Device EEPROM O.K.
	DATA	1		User
	DATA	3	0x0	NA